Post-hoc analyses of the cross-validated copy-paste experiment

Error analysis, detection versus segmentation, and augmentation versus model capacity

EGH490 – analysis notes (12 October 2026)

0. Data, validity and common methods

- **No retraining or new inference.** All analyses use the test-fold predictions saved by the 120 finished models (6 set-ups $\times$ 2 arms $\times$ 5 folds $\times$ 2 seeds): every box with confidence ≥ 0.001 (top-1 class, NMS IoU 0.7, $\leq$300 per frame), and for segmentation the mask IoU of every prediction with every labelled fish. A CPU-only job re-packed these files, the ground truth, the training logs and the model sizes. The saved predictions contain only each box's top class, so second-choice species cannot be analysed.
- **Same partitions, no leakage.** Each frame is scored only by the model whose test fold contains it (checked for every prediction), using `folds.csv` and the ground-truth files of `cv_evaluate.py`. Re-computing the published values from the re-packed predictions reproduced them exactly (e.g. medium detection focus-species AP50 0.335/0.424; medium segmentation mask AP50 0.308/0.458). A conservative check found that 7 of the 203 focus fish (all Orangespine Unicornfish, in two long encounters) have other frames of the same encounter in the training/validation pool of the model that scores them; all other focus fish are from encounters entirely outside that pool.
- **Uncertainty.** Paired block bootstrap: the 49 blocks of 30 frames resampled 2,000 times with the generator and seed used by `cv_evaluate.py`; the same resample is applied to every model, so baseline/copy-paste, detection/segmentation and small/medium/large comparisons are all paired by block. Seeds are averaged within each resample. Intervals are 95% percentile intervals; p is the two-sided bootstrap p. Seed-to-seed variance is *not* included.
- **Status.** These analyses were *not* pre-registered. They are exploratory and the intervals are not adjusted for multiple comparisons; they should be reported as secondary and descriptive.

1 Error analysis of the three focus species

Operating point and definitions. Confidence threshold $\tau = 0.25$ (the threshold of the reported precision and recall); box IoU threshold 0.5 for all set-ups (so detection and segmentation are compared on the same criterion). Predictions with confidence $\geq \tau$ are matched to labelled fish of the same class, highest confidence first (as in COCO/`cv_evaluate.py`). Each labelled focus fish then falls into exactly one category, checked in this order:

(a) *Correct species* (TP): matched as above.

(b) *Wrong species* (classification error): an unmatched prediction of another species with IoU ≥ 0.5 and confidence $\geq \tau$ (predictions already matched to a neighbouring fish are excluded).

(c) *Poorly localised* (localisation error): an unmatched prediction with $0.1 \leq$ IoU < 0.5 and confidence $\geq \tau$.

(d) *Missed*: no prediction with IoU ≥ 0.1 and confidence $\geq \tau$. Missed fish are further split by whether a *correct-species* box with IoU ≥ 0.5 exists below τ. These sub-threshold scores are reported only as evidence of a weak signal, never as classifications (their median confidence is 0.02).

False positives (unmatched focus-species predictions, confidence $\geq \tau$) are split into duplicates (IoU ≥ 0.5 with an already-matched fish of that species), wrong-species FPs (IoU ≥ 0.5 with a labelled fish of another species), poorly localised FPs, and FPs on *unlabelled regions* (IoU < 0.1 with every labelled fish). The last group may contain real but unlabelled fish; it was not re-annotated and should not be read as hallucination. Results below pool all six set-ups and both seeds (12 models per arm; per-set-up results are in `components_by_setup.csv` and `err_summary_all.csv`).

Table 1: Fate of every labelled focus fish, baseline → copy-paste (pooled over 12 models per arm; 93, 77 and 33 fish). Second line: change with 95% block-bootstrap interval; bold: interval excludes zero. pp = percentage points.

Quantity	Forcepsfish	Orangespine Unicornfish	Convict Tang
Recall (%)	26.8 → 40.3	15.5 → 19.6	35.6 → 58.8
	+13.5 [+6.8, +23.2]	**+4.1 [+1.6, +9.3]**	**+23.2 [+16.8, +34.9]**
Precision (%)	57.7 → 60.3	36.3 → 41.4	59.0 → 53.3
	+2.6 [−3.0, +10.8]	+5.1 [−0.3, +13.2]	−5.7 [−13.1, +7.1]
Wrong species (%)	24.7 → 17.2	15.4 → 14.2	11.6 → 7.6
	−7.5 [−14.3, −2.1]	−1.2 [−3.8, +3.1]	**−4.0 [−8.3, −1.5]**
Poorly localised (%)	8.2 → 5.6	2.7 → 1.8	0.3 → 0.3
	−2.6 [−5.0, −0.4]	−0.9 [−2.6, +0.2]	+0.0 [+0.0, +0.0]
Missed, no prediction ≥0.25 (%)	40.2 → 36.8	66.5 → 64.4	52.5 → 33.3
	−3.4 [−8.2, +0.2]	−2.1 [−10.5, +1.6]	**−19.2 [−28.3, −14.3]**
of which correct species <0.25 (%)	25.8 → 20.7	37.1 → 31.9	32.1 → 21.2
	−5.1 [−9.5, −1.1]	**−5.2 [−13.5, −1.6]**	−10.9 [−25.9, +6.9]
Localised, any species (%)	51.5 → 57.5	30.8 → 33.8	47.2 → 66.4
	+6.0 [+2.6, +10.8]	+2.9 [−1.6, +10.8]	**+19.2 [+14.3, +28.3]**
Correct species | localised (%)	52.0 → 70.1	50.2 → 58.0	75.4 → 88.6
	+18.1 [+7.6, +33.8]	+7.8 [−1.1, +14.2]	**+13.2 [+3.9, +37.2]**
FP on other species (per model)	6.6 → 11.2	7.8 → 6.5	4.0 → 9.6
	+4.7 [+1.0, +9.4]	−1.2 [−3.3, +0.8]	**+5.6 [+1.2, +11.8]**
FP on unlabelled regions (per model)	7.8 → 9.4	12.6 → 14.3	3.2 → 6.9
	+1.6 [−1.7, +4.7]	+1.8 [−1.8, +5.4]	**+3.7 [+1.5, +6.2]**
All FP (per model)	18.2 → 24.7	20.9 → 21.3	8.2 → 17.0
	+6.4 [+0.7, +13.0]	+0.4 [−3.9, +4.4]	**+8.8 [+3.9, +15.4]**

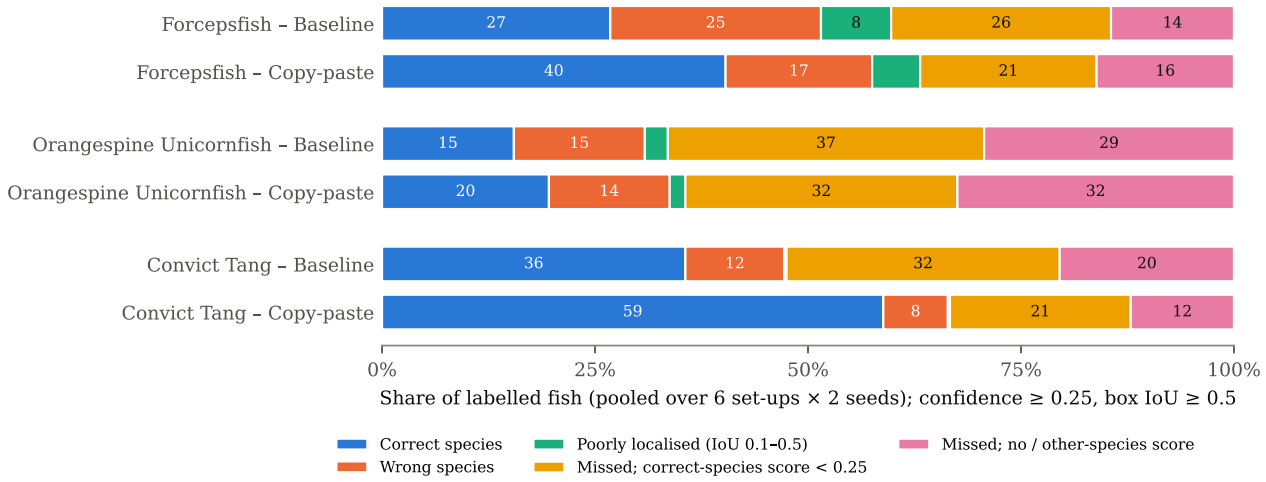


Figure 1: What happened to every labelled focus fish, by species and arm (confidence ≥ 0.25, box IoU ≥ 0.5).

Table 2: Most frequent species confusions (counts per model, mean of 12 models; confidence ≥ 0.25, IoU ≥ 0.5). †Class spelt "Lognnose" in the label set.

Species	Direction	Baseline	Copy-paste
Forcepsfish	Labelled as (FN)	Yellow Tang 22.7; Lined Butterflyfish 0.1	Yellow Tang 15.6; Convict Tang 0.2
	Other fish labelled as it (FP)	Yellow Tang 3.2; Longnose Butterflyfish† 1.8	Yellow Tang 5.0; Longnose Butterflyfish† 2.2
Orangespine Unicornfish	Labelled as (FN)	Brown Surgeonfish 6.8; Goldring Surgeonfish 1.2	Brown Surgeonfish 6.0; Moorish Idol 1.2
	Other fish labelled as it (FP)	Brown Surgeonfish 1.3; Acanthuridae 1.2	Brown Surgeonfish 1.8; Acanthuridae 1.2
Convict Tang	Labelled as (FN)	Ornate Butterfly Fish 1.5; Yellow Tang 1.2	Ornate Butterfly Fish 0.8; Yellow Tang 0.7
	Other fish labelled as it (FP)	Ornate Butterfly Fish 1.3; Indo-Pacific Sergeant 1.0	Ornate Butterfly Fish 5.2; Multiband Butterflyfish 1.2

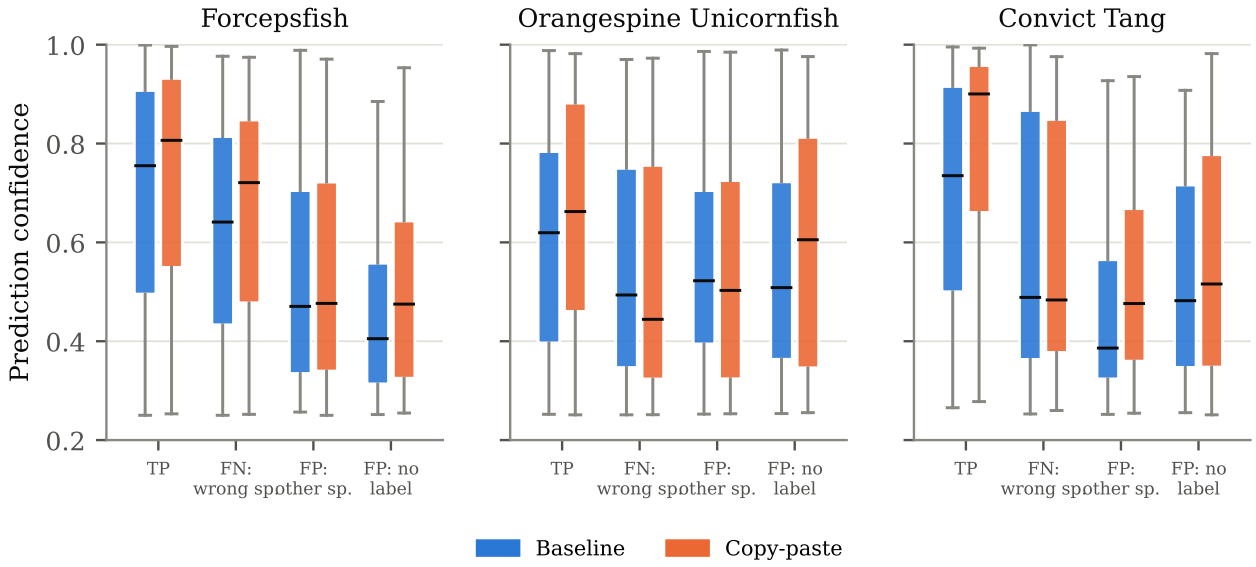


Figure 2: Confidence of correct detections (TP), of wrong-species labels given to focus fish (FN), and of focus-species false positives on other species or on unlabelled regions. All predictions shown have confidence ≥ 0.25 by construction.

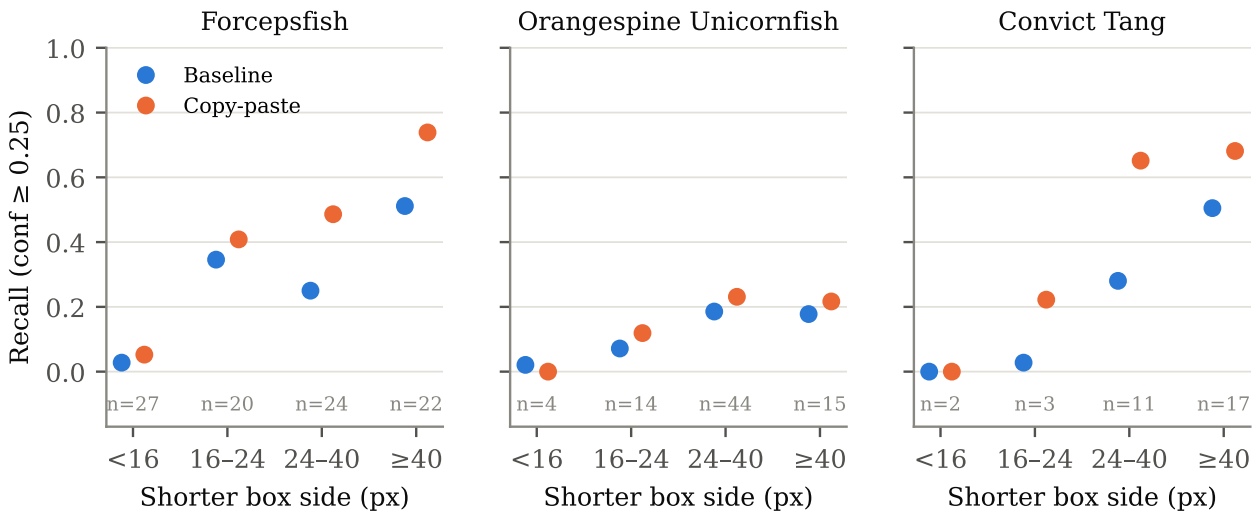


Figure 3: Recall by fish size (shorter box side), baseline and copy-paste; n = labelled fish per bin. Bins with few fish (e.g. Convict Tang below 24 px) are not interpretable.

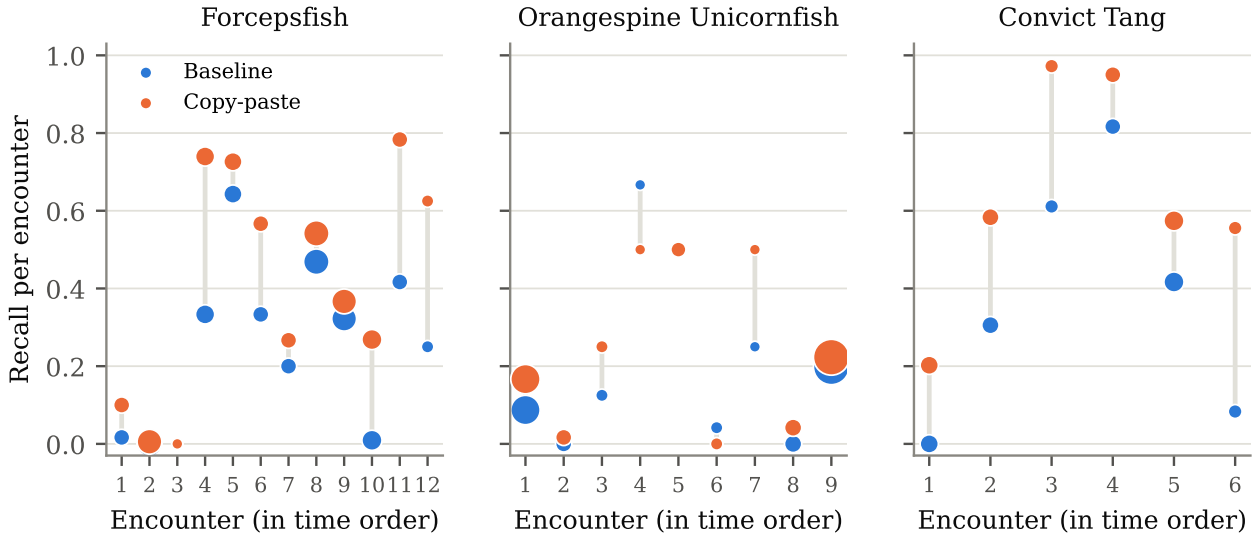


Figure 4: Recall per encounter (runs of frames ≤15 apart), baseline to copy-paste; marker area ∝ fish in the encounter.

Interpretation

- **Baseline failures.** For all three species the largest failure category is a fish with no prediction at confidence ≥ 0.25 (Forcepsfish 40%, Orangespine 67%, Convict 52% of fish). Wrong-species labels are the second category (25%, 15%, 12%); poor localisation is small (8%, 3%, 0%). More than half of the missed fish (56–64%) have a correct-species box below the threshold, but with very low confidence (median 0.02).
- **How copy-paste helped differed by species.** Forcepsfish gained mainly through fewer wrong-species labels (−7.5 pp) and fewer poorly localised boxes (−2.6 pp); the missed share did not change significantly. Convict Tang gained mainly through fewer missed fish (−19.2 pp), with a smaller drop in wrong-species labels (−4.0 pp). Orangespine Unicornfish changed little (recall +4.1 pp; no category changed significantly).
- **Confusions.** Most Forcepsfish classification errors are Yellow Tang (22.7 of 93 fish per model at baseline, 15.6 with copy-paste). Orangespine Unicornfish is mostly confused with Brown Surgeonfish, and Convict Tang with Ornate Butterflyfish and Yellow Tang.

- **Cost in false positives.** Copy-paste raised focus-species false positives (pooled 47 → 63 per model), mainly by labelling other fish as Forcepsfish or Convict Tang (e.g. Ornate Butterflyfish labelled Convict Tang: 1.3 → 5.2 per model) and by more predictions on unlabelled regions. Precision did not change significantly for any species (Convict Tang −5.7 pp, interval −13.1 to +7.1).
- **Confidence.** Correct detections became more confident with copy-paste (median Convict Tang 0.74 → 0.90, Forcepsfish 0.76 → 0.81). Wrong-species labels and false positives have overlapping confidence distributions, so no single threshold separates them.
- **Size and encounters.** Recall increases with fish size for every species and arm. For Forcepsfish below 16 px recall is 0.03 (baseline) and 0.05 (copy-paste); the gains are concentrated in fish of 24 px and larger. Copy-paste improved recall in 10 of 12 Forcepsfish encounters (none worse), all 6 Convict Tang encounters, and 5 of 9 Orangespine encounters (2 worse). One Forcepsfish encounter (15 fish, median 12 px) stays near zero under both arms.

Supported by the data versus hypotheses

Supported: the category shares and their changes in Table 1 (within this footage, these partitions and this operating point); the dominant confusion pairs; that gains are concentrated in larger fish and spread over most encounters of Forcepsfish and Convict Tang; that false positives increased while precision did not change significantly. **Hypotheses only:** *why* Forcepsfish is confused with Yellow Tang (e.g. shared colour) and *why* copy-paste reduces that confusion (e.g. more varied examples) were not tested; whether predictions on unlabelled regions are real fish; whether tiny fish are missed because of image resolution. These are correlational observations, not causal mechanisms. Results depend on the chosen thresholds; a different τ moves fish between "missed" and "correct".

2 Does copy-paste help segmentation more than detection?

The pre-registered tests used box AP50 for detection and mask AP50 for segmentation, so they cannot show that the improvement magnitudes differ. Here the copy-paste effect (Δ = copy-paste − baseline, seed-averaged) is computed for every model on *both* box and mask metrics, and the *difference in effects* between tasks at matched size is bootstrapped with the same block resamples for both tasks (difference-in-differences).

Table 3: Copy-paste effect on focus-species mean AP, by task and size, and the difference in effects (segmentation box − detection box; p in brackets). Last column: within segmentation, mask effect − box effect.

Metric	Size	Det. (box)	Seg. (box)	Seg. (mask)	Seg. box − det. box	Seg. mask − box
AP50	Small	**+0.090 [+0.026, +0.150]**	**+0.112 [+0.040, +0.176]**	**+0.116 [+0.049, +0.176]**	+0.022 [−0.041, +0.085] (0.491)	+0.005 [−0.004, +0.014]
AP50	Medium	**+0.089 [+0.041, +0.144]**	**+0.148 [+0.095, +0.218]**	**+0.149 [+0.099, +0.217]**	**+0.060 [+0.013, +0.110]** (0.026)	+0.001 [−0.004, +0.007]
AP50	Large	**+0.068 [+0.015, +0.138]**	**+0.089 [+0.026, +0.153]**	**+0.084 [+0.020, +0.148]**	+0.022 [−0.063, +0.091] (0.542)	−0.006 [−0.013, +0.000]
AP50	Mean of sizes	–	–	–	+0.034 [−0.007, +0.072] (0.100)	−0.000 [−0.005, +0.004]
AP50-95	Small	**+0.071 [+0.022, +0.124]**	**+0.094 [+0.044, +0.144]**	**+0.087 [+0.036, +0.138]**	+0.023 [−0.025, +0.070] (0.326)	−0.007 [−0.015, +0.002]
AP50-95	Medium	**+0.079 [+0.037, +0.123]**	**+0.121 [+0.076, +0.181]**	**+0.117 [+0.077, +0.169]**	**+0.042 [+0.005, +0.085]** (0.032)	−0.004 [−0.015, +0.007]
AP50-95	Large	**+0.049 [+0.006, +0.099]**	**+0.085 [+0.030, +0.140]**	**+0.073 [+0.021, +0.125]**	+0.036 [−0.021, +0.094] (0.238)	−0.012 [−0.025, +0.002]
AP50-95	Mean of sizes	–	–	–	**+0.034 [+0.000, +0.066]** (0.049)	**−0.007 [−0.013, −0.001]**

Table 4: Difference in copy-paste effect per species, segmentation box AP50 − detection box AP50.

Species	Small	Medium	Large	Mean of sizes
Forcepsfish	+0.053 [−0.029, +0.125]	+0.068 [−0.007, +0.149]	−0.015 [−0.112, +0.088]	+0.035 [−0.018, +0.086]
Orangespine Unicornfish	−0.047 [−0.129, +0.088]	+0.015 [−0.059, +0.135]	+0.072 [−0.100, +0.168]	+0.013 [−0.035, +0.064]
Convict Tang	+0.059 [−0.118, +0.217]	**+0.096 [+0.012, +0.180]**	+0.009 [−0.124, +0.162]	+0.054 [−0.041, +0.146]

Table 5: Decomposition of the copy-paste effect for the pooled focus fish (confidence ≥ 0.25). Localisation: fish with an any-species box at IoU ≥ 0.5. Identification: correct species among localised fish. Cells: detection effect / segmentation effect, then segmentation − detection with 95% interval.

Component	Small	Medium	Large
Recall (pp)	+7.9 / +10.6	+11.3 / +15.5	+11.3 / +12.6
	+2.7 [−3.5, +9.7]	+4.2 [−2.3, +11.5]	+1.2 [−9.0, +9.6]
Localisation (pp)	+10.6 / +2.5	+5.9 / +11.3	+1.2 / +10.3
	−8.1 [−16.4, −0.0]	+5.4 [−1.7, +12.6]	+9.1 [−0.3, +16.2]
Identification (pp)	+4.1 / +21.5	+15.6 / +16.8	+21.5 / +11.1
	+17.4 [+7.5, +28.8]	+1.1 [−9.2, +12.2]	**−10.5 [−18.8, −2.4]**
Mask IoU of correct detections	0.844 → 0.854	0.858 → 0.861	0.864 → 0.860
(seg only)	+0.010 [−0.029, +0.038]	+0.003 [−0.027, +0.023]	−0.003 [−0.022, +0.012]

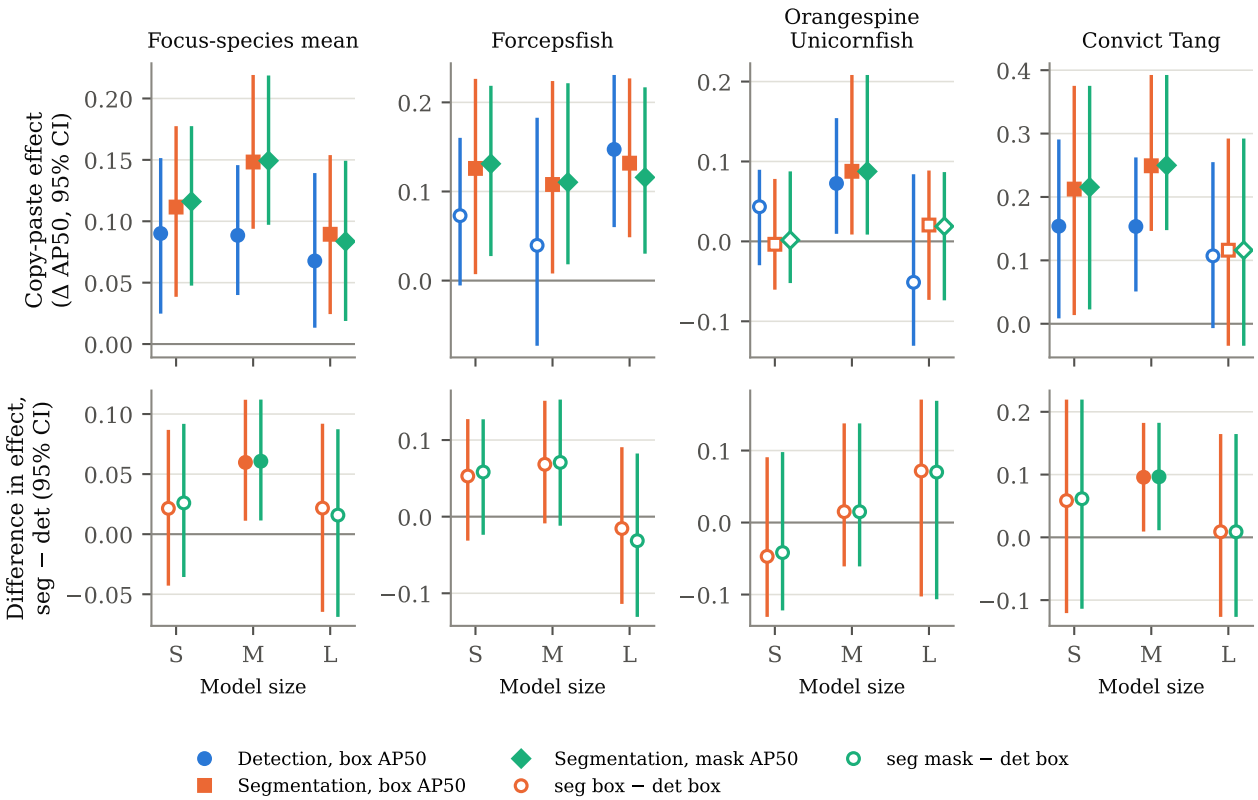


Figure 5: Top: copy-paste effect (AP50) for detection (box) and segmentation (box and mask) at each size. Bottom: difference in effect, segmentation − detection. Filled markers: interval excludes zero.

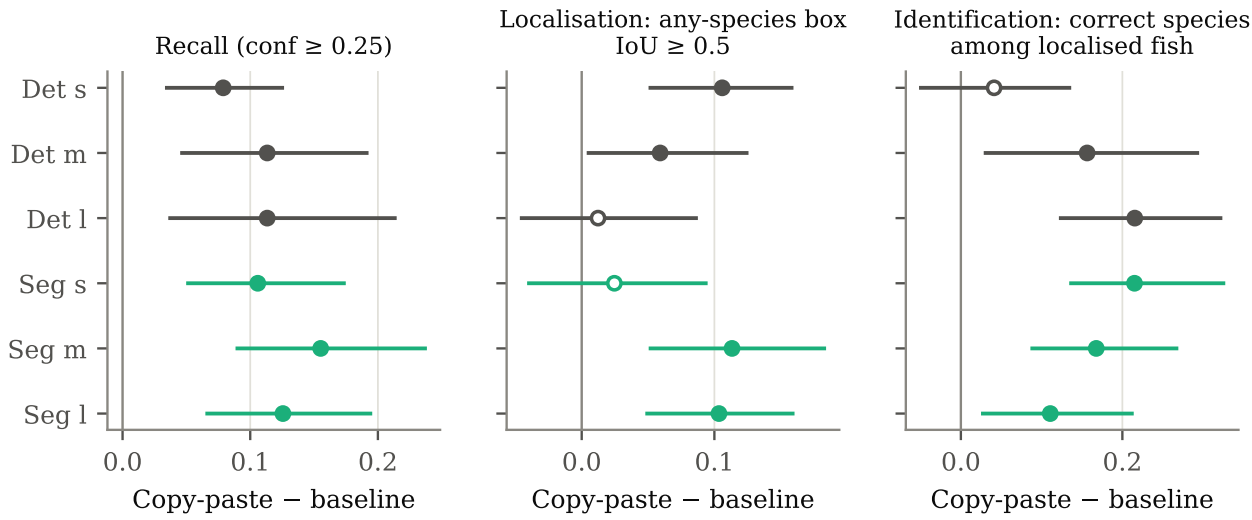


Figure 6: Copy-paste effect on recall, localisation and identification of focus fish for each set-up.

Interpretation

- **Metric choice does not explain the apparent difference.** Within segmentation, the mask-AP50 effect equals the box-AP50 effect (mean difference -0.000, interval -0.005 to $+0.004$). Using mask AP for segmentation therefore did not inflate its effect at IoU 0.5; at AP50-95 the mask effect is slightly *smaller* than the box effect (-0.007).
- **The task difference is not robust.** On the same metric (box AP50), the segmentation effect exceeds the detection effect by $+0.022$ (small), $+0.060$ (medium) and $+0.022$ (large). Only the medium difference has an interval excluding zero ($+0.013$ to $+0.110$, $p = 0.026$), which would not survive a correction for three sizes (Bonferroni $p \approx 0.08$); averaged over sizes the difference is $+0.034$ (-0.007 to $+0.072$, $p = 0.10$). No single species shows a consistent difference.
- **Components.** The routes by which copy-paste raised recall differ between set-ups without a consistent task pattern: at small size segmentation gained more in identification but less in localisation than detection; at large size the opposite; at medium neither differed. Net recall differences are not significant at any size.
- **Mask quality.** Copy-paste did not measurably change mask quality: the mask IoU of correctly detected focus fish was unchanged at every size (changes -0.003 to $+0.010$, all intervals include zero), and every box-matched segmentation detection had mask IoU ≥ 0.60 (median 0.88). Mask quality is not the limiting factor for these species.

Conclusion. The evidence does not support a task-dependent augmentation effect. Both tasks benefit, by similar amounts once the same metric is used; the larger numbers for segmentation in the primary table come mainly from one size (medium) and lie within the statistical uncertainty. Segmentation should not be described as benefiting more. Caveats: detection and segmentation are different networks trained on different label formats, two seeds only, and seed variance is not in the intervals.

3 Targeted augmentation versus model capacity

Table 6: Engineering trade-off. Parameters and GFLOPs from one trained checkpoint per architecture (58-class head), GFLOPs at the 1280-px input used in training and testing. Checkpoint: `best.pt` (FP16, optimiser stripped). Inference: median per-image GPU time during the batch-8 test validation on an NVIDIA H100 (Ultralytics "inference" timer, excluding pre- and post-processing) – a throughput-style measurement on a data-centre GPU, *not* a latency or edge-device benchmark. Training: mean wall-clock training loop per run; GPU-h: sum of PBS wall time over the 10 runs (one H100 each). AP50: box, pooled CV, mean of 2 seeds.

Model	Arm	Params (M)	GFLOPs	Ckpt (MB)	H100 ms/img	Train min/run	GPU-h (10 runs)	Focus AP50	Non-target AP50[a]	Any-fish AP50
Det S	Base	10.0	95	20	1.9	19	3.5	0.267	0.262	0.670
	CP	10.0	95	20	1.9	24	4.4	0.357	0.257	0.667
Det M	Base	21.9	305	44	3.2	24	4.4	0.335	0.296	0.694
	CP	21.9	305	44	3.2	40	7.1	0.424	0.295	0.687
Det L	Base	26.3	380	53	4.2	38	6.5	0.373	0.294	0.681
	CP	26.3	380	53	4.1	57	9.7	0.440	0.310	0.672
Seg S	Base	11.5	153	24	2.4	25	4.4	0.233	0.256	0.668
	CP	11.5	153	24	2.3	40	6.8	0.345	0.260	0.658
Seg M	Base	27.1	533	55	4.1	41	7.0	0.303	0.274	0.683
	CP	27.1	533	55	4.1	50	8.4	0.451	0.299	0.693
Seg L	Base	31.5	609	64	4.9	53	9.2	0.351	0.290	0.671
	CP	31.5	609	64	4.9	76	13.1	0.441	0.308	0.677

[a] Mean AP50 over the 17 well-labelled ($\geq$20 instances) classes that were not augmented.

Table 7: Paired comparisons (box AP50, block bootstrap; bold: interval excludes zero).

Task	Comparison	Focus-species mean	Non-target (17 classes)	Any fish
Detection	CP small − baseline medium	+0.022 [−0.040, +0.088]	**−0.039 [−0.077, −0.001]**	**−0.026 [−0.039, −0.012]**
Detection	CP medium − baseline large	**+0.051 [+0.011, +0.107]**	+0.001 [−0.014, +0.029]	+0.006 [−0.013, +0.023]
Detection	Augmentation gain (small)	**+0.090 [+0.026, +0.150]**	−0.005 [−0.029, +0.031]	−0.003 [−0.016, +0.012]
Detection	Capacity gain small→medium	**+0.068 [+0.014, +0.114]**	**+0.034 [+0.014, +0.063]**	**+0.023 [+0.013, +0.035]**
Detection	Augmentation gain (medium)	**+0.089 [+0.041, +0.144]**	−0.001 [−0.016, +0.021]	−0.007 [−0.019, +0.005]
Detection	Capacity gain medium→large	+0.037 [−0.015, +0.090]	−0.002 [−0.027, +0.013]	−0.013 [−0.025, +0.000]
Segmentation	CP small − baseline medium	+0.042 [−0.007, +0.103]	−0.014 [−0.033, +0.003]	**−0.025 [−0.044, −0.005]**
Segmentation	CP medium − baseline large	**+0.100 [+0.048, +0.174]**	+0.009 [−0.015, +0.039]	**+0.022 [+0.007, +0.033]**
Segmentation	Augmentation gain (small)	**+0.112 [+0.040, +0.176]**	+0.004 [−0.011, +0.021]	−0.010 [−0.023, +0.004]
Segmentation	Capacity gain small→medium	**+0.070 [+0.011, +0.109]**	**+0.019 [+0.002, +0.038]**	+0.015 [−0.002, +0.033]
Segmentation	Augmentation gain (medium)	**+0.148 [+0.095, +0.218]**	**+0.024 [+0.006, +0.049]**	+0.010 [−0.007, +0.026]
Segmentation	Capacity gain medium→large	+0.048 [−0.009, +0.104]	+0.016 [−0.004, +0.035]	−0.012 [−0.026, +0.005]

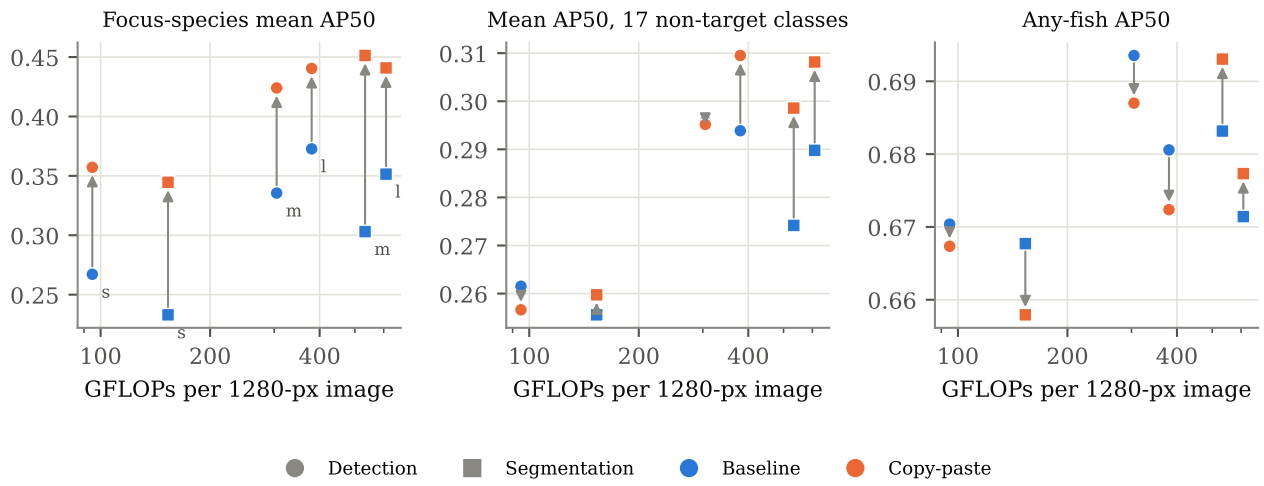


Figure 7: Performance against inference compute (GFLOPs per 1280-px image, log scale) for the baseline and copy-paste version of each model; arrows show the copy-paste change. Augmentation does not change inference compute.

Interpretation

- **Inference cost.** Copy-paste leaves architecture, parameters, FLOPs, checkpoint size and measured H100 inference time unchanged; moving up one size increases GFLOPs by 1.1–3.5× (e.g. detection 95 → 305 → 380).
- **Training cost.** Copy-paste adds 450 synthetic images to about 820 real ones, raising time per epoch by about 50% and training time per run by 22–67% (runs also stop at different epochs) (e.g. medium detection 24 → 40 min). Generating the synthetic images took 2–5 minutes of CPU time. The whole 120-run experiment used 84.6 H100 GPU-hours.
- **Augmented small vs baseline medium.** For the focus species the augmented small models are statistically indistinguishable from the baseline medium models (detection +0.022, −0.040 to +0.088; segmentation +0.042, −0.007 to +0.103), but they are worse on non-target species (detection −0.039) and on finding fish at all (any-fish −0.026 detection, −0.025 segmentation). The augmented small model is not a substitute for the medium model.
- **Augmented medium vs baseline large.** The augmented medium models outperform the baseline large models on the focus species (detection +0.051, +0.011 to +0.107; segmentation +0.100, +0.048 to +0.174), with no loss on non-target species or any-fish AP50, at 80–88% of the large model's FLOPs.
- **Augmentation versus capacity for under-represented species.** At medium size the augmentation gain exceeds the medium-to-large capacity gain (detection by +0.051, segmentation by +0.100, both intervals exclude zero); at small size the two gains are not distinguishable. Capacity, unlike augmentation, also improves the non-target species (small → medium: +0.034 detection, +0.019 segmentation), whereas augmentation's effect on non-target species is near zero.
- **No harm to non-target species.** Augmentation did not significantly reduce non-target AP50 or any-fish AP50 in any of the six set-ups (largest decreases −0.005 and −0.010, intervals include zero).

Conclusion. For a deployment that must keep a given model size, targeted augmentation improves the under-represented species at no inference cost and a modest one-off training cost, and at medium size it is a better route to those species than moving to the large model. It is not a replacement for adding capacity when overall species identification or fish detection matters, and none of these numbers describe performance on an edge device, which was not benchmarked.

4 Implications for the report

- Replace "copy-paste gave a gain similar to moving up one model size" with the paired results: comparable to (not significantly different from) small→medium for the target species only, and significantly better than medium→large. The current sentence "small detection with copy-paste (0.357) exceeded medium detection without it (0.335)" compares point estimates; the difference is not significant.
- Do not state that segmentation benefits more; state that both tasks benefit by similar amounts on the same metric.
- Revise "most missed Forcepsfish are predicted as background": at $\tau = 0.25$, 55% of the Forcepsfish not correctly detected at baseline had no prediction, 34% were labelled another species (almost all Yellow Tang) and 11% were poorly localised.
- TODOs that can now be filled: optimiser AdamW (Ultralytics "auto", lr 0.000161) in all 120 runs; AutoBatch sizes 14/8/6 (detection s/m/l) and 12/6/5 (segmentation); Ultralytics 8.4.126, PyTorch 2.13.0; 84.6 H100 GPU-hours (energy upper bound about 59 kWh at the 700 W board power, not measured).